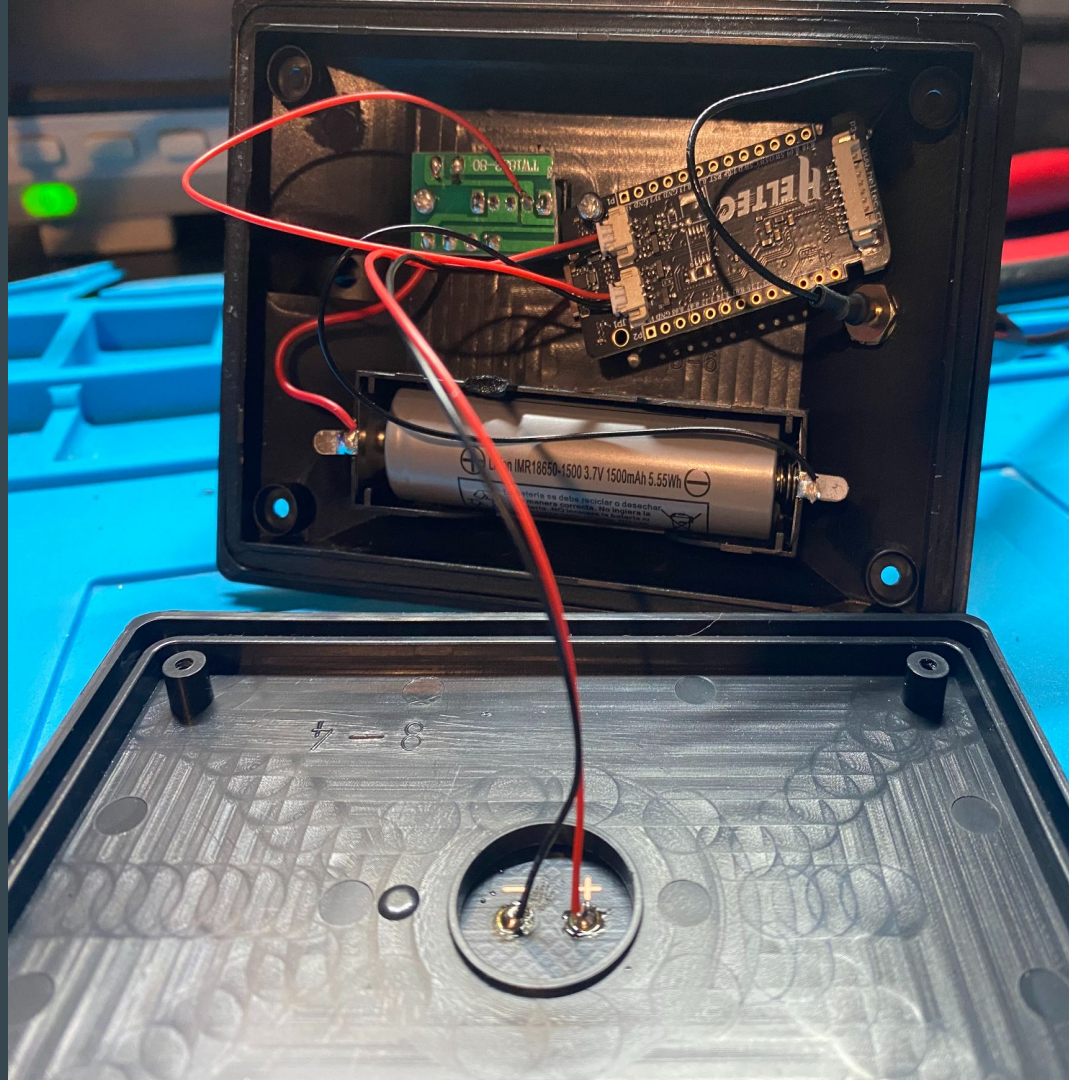


meshtastic essentials

a free workshop held on
2026 mm xx
at Prototype North Oakland
by the Protomesh Collective



How much do you already know?

- 0 "Mesh-what?"
- 1 Heard the name
- 2 Read about it
- 3 Own a device
- 4 Sending messages
- 5 Building solar nodes

SLIDE SHAMELESSLY STOLEN FROM SUNBEAR

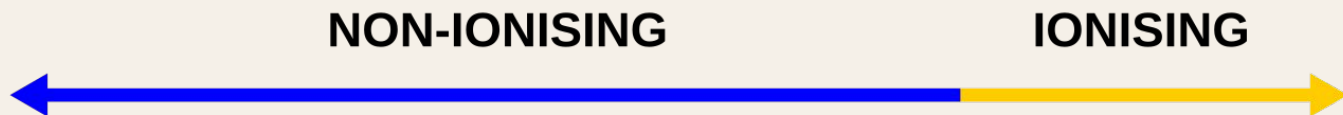
what do you think Meshtastic is?

shout it out! no wrong answers

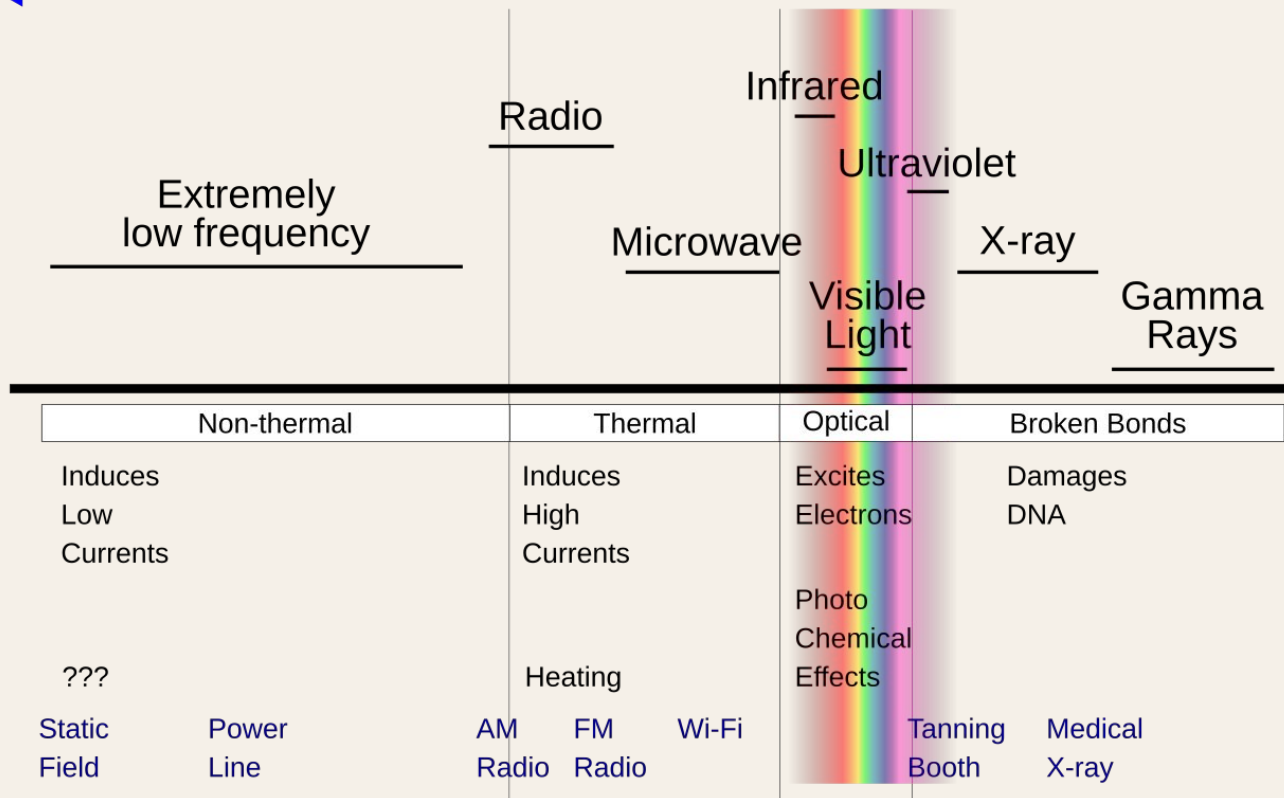
SLIDE SIMILARLY STOLEN FROM SUNBEAR

What is Meshtastic?

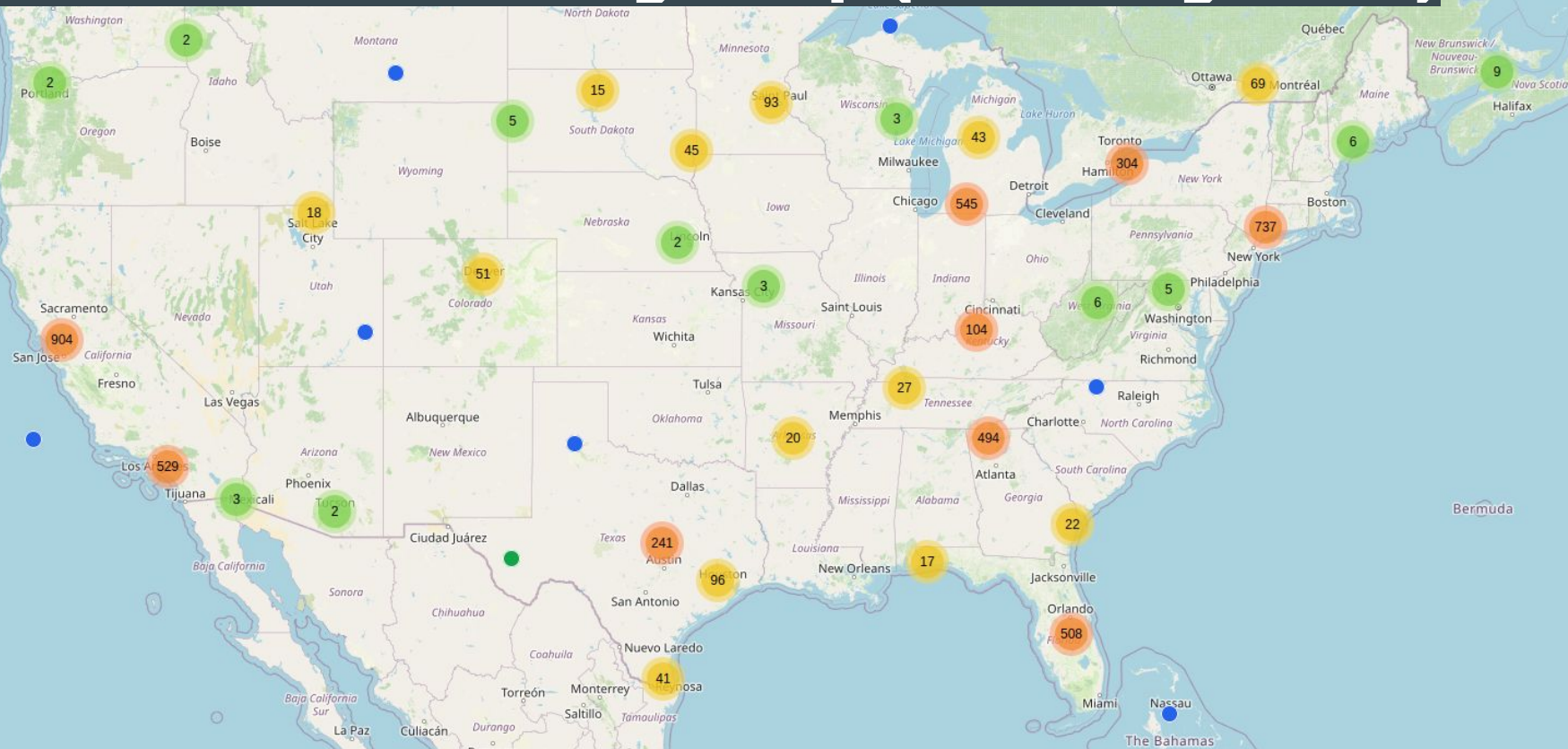
- Free and open source communications technology
 - Text messaging over radio
 - Decentralized, encrypted, and hyperresilient
- The nodes themselves are the network
 - Messages hop from node to node
 - More nodes increase network coverage
- Uses LoRa (“LOng RAnge”) radio waves
 - 900-930 MegaHertz band centered on 915 MHz (33cm wavelength)
 - ½ to 1 mile of range with stock antenna



The Electromagnetic Spectrum (from Wikipedia)



United States Coverage Map (February 2026)



United States Coverage Map (August 2026)

The map displays coverage data for the United States and surrounding regions. The data is represented by colored circles with numbers inside, indicating coverage levels. Blue location pins are also present across the map. The map includes state and city names, and a legend in the bottom right corner.

Legend:

- 9,429 nodes from Meshtastic MQTT
- Map tiles from OpenStreetMap

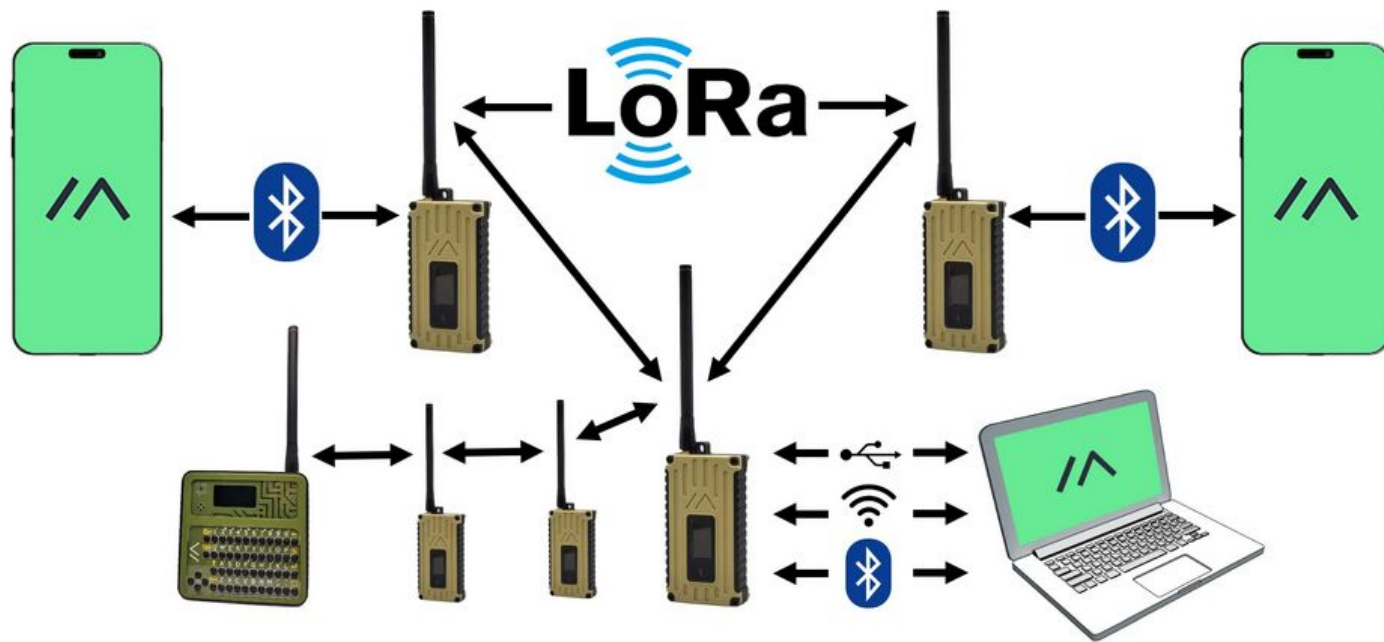
United States Coverage Map (August 2026)

The map displays coverage data for the United States and parts of Mexico. The data is represented by colored circles (yellow, orange, red, green) and blue location pins. Each circle contains a number, likely representing a coverage metric. The map shows a high density of coverage points in the eastern half of the United States, particularly in the Northeast and Southeast, and a lower density in the western half. The map also shows the Gulf of Mexico and the Caribbean Sea.

9,429 nodes from Meshtastic MQTT | Map tiles from OpenStreetMap

Why Meshtastic?

- Hyperresilient - it works when nothing else does
 - Doesn't need cell signal, Internet, or even the power grid
 - Ideal for disaster recovery AND remote sensor deployments
- Decentralized and community-driven
 - Don't need to wait for or pay for centralized network infrastructure
 - Volunteer/open-source projects bring out the best of people and their skills
 - "Pittsburgh has the best Mesh we've seen" - Anthrocon 2026
- End-to-end encrypted
 - Meshtastic's cryptosystem is NOT as good as Signal's!
 - But, Signal gets to assume a reliable network, and ham radio doesn't allow encryption at all
 - Encryption is convivial technology that enables experimental projects to become practical comms



How does the Meshtastic network work?

(US) Meshtastic nodes talk to each other over (915 MHz) LoRa radio waves. We usually talk to nodes via Bluetooth, but we can also use USB or sometimes Wifi.

How do Meshtastic messages travel?

1. Your phone pairs to the device over bluetooth
2. You send your text message to the device
3. The device sends the message to the network over LoRa
4. The nodes in the network repeat the message a certain number of hops, hopefully reaching the recipient
5. Their device sends the message over Bluetooth to their phone



Public-Private Key Cryptography

- Asymmetric encryption generates a pair of two keys
 - Public key -> Shared publicly, associated with your online persona
 - Private key -> DO NOT SHARE, Anyone with this key can decrypt all of your messages
- Encrypt with Public Key -> Decrypt with Private Key
(Secure Comms)
- Encrypt with Private Key -> Decrypt with Public Key
(Digital Signatures)
- Asymmetrical cryptography allows you to securely send and receive messages, simply by knowing each others public keys.

Cryptography Examples

- You encrypt a message with someone else's public key and send it to a server. Only they can decrypt that message with their private key.
- You encrypt a message with your private key and send it to a server. Anyone with your public key can decrypt the message, and knows that you're the only person who could have made it.
- You encrypt a message with BOTH your private key and someone else's public key. You both know that you are the only ones who can read the message. They know that you must have sent it, not an imposter.

There Is No Perfect Technology

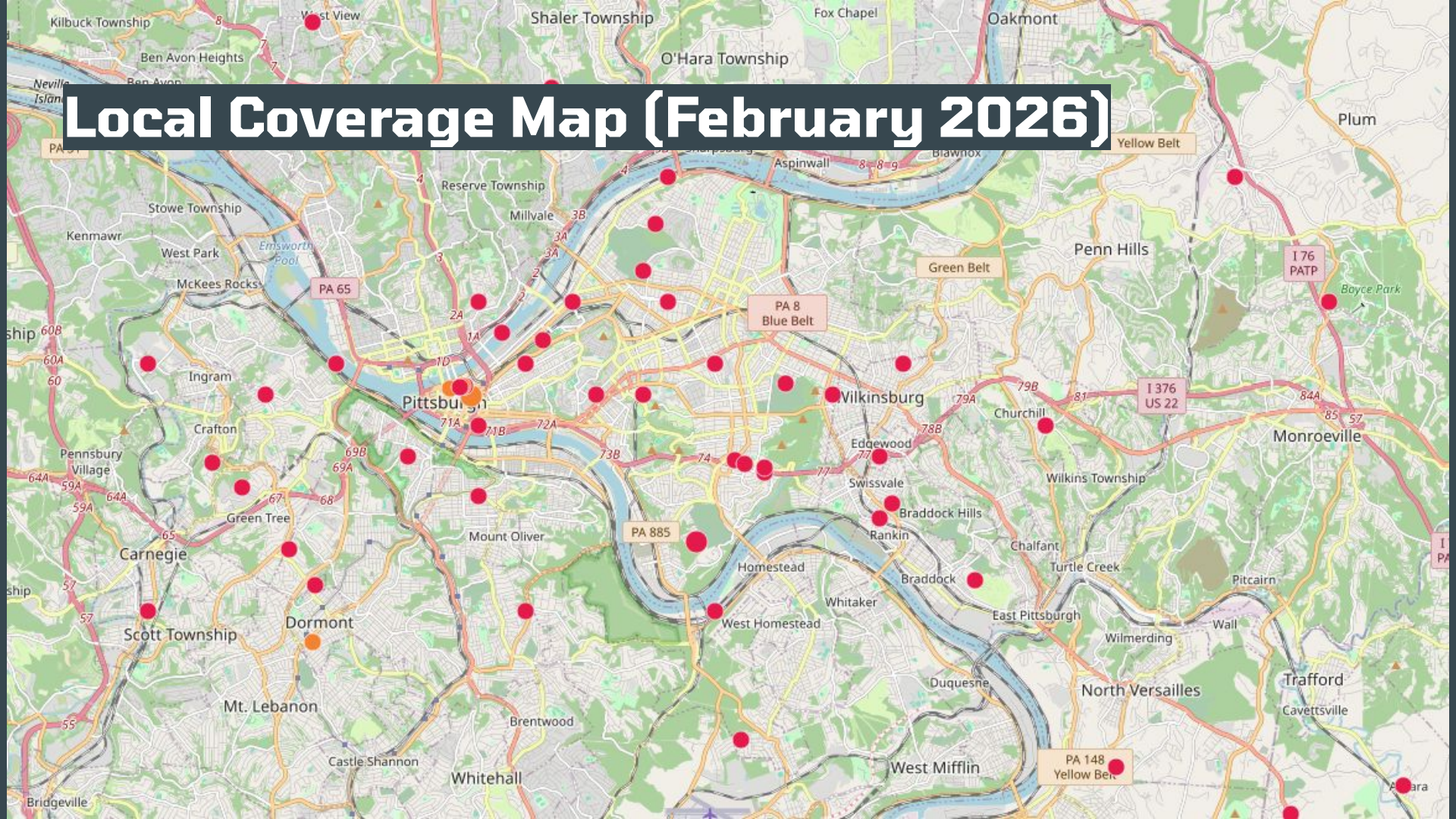
Technology Name	Technology Description	Pros	Cons	Best For
LoRa	Method for encoding data in radio waves of particular frequencies (uses 915 MHz in US, 868 MHz in Europe)	Uses frequency bands that don't require operator licensing, good at connecting hard-to-reach locations	Low bandwidth means slow data speeds	Enabling "meshnet" projects such as Meshtastic and Meshcore
Meshtastic	LoRa-based decentralized encrypted text comms (easy to use)	Extremely easy to set up, works well when few nodes around	Transmits data quite inefficiently	Building a LoRa meshnet where none exists
Meshcore	LoRa-based decentralized encrypted text comms (efficient)	Faster, more reliable, longer range than Meshtastic	More difficult to configure, needs dedicated routing nodes AND client nodes	Making an established LoRa meshnet more efficient
Signal	End-to-end encrypted text and voice app	Most secure cryptosystem accessible to the average user	Needs Internet and/or cell network to operate	Transmitting info that needs to stay encrypted for 2+ years*
Ham Radio	Hyperresilient voice/data comms on other frequency bands	Gear more powerful than Meshtastic, can transmit more info faster, long-established community	Requires licensing, encryption forbidden, gear more expensive	Assisting official/state-funded disaster recovery efforts

...but we keep getting closer!

Real-World Examples

- Asheville, North Carolina
 - After Hurricane Helene destroyed large chunks of Asheville's infrastructure, people were able to use combinations of Meshtastic and GMRS radios to spread information when the cell towers were out. There are still 65 active nodes in the Asheville portion of the North Carolina Mesh Network.
- Redtail Resilience Network
 - Enabling both hyperlocal community organization in Pittsburgh and disaster-recovery volunteer coordination from East Palestine to New Orleans
 - Led in part by Sunbear, who also helps lead WPAmesh
- Ukraine
 - Meshtastic is being used for both combat and non-combat roles where access to encrypted radios is limited. Meshtastic pairs well with ATAK systems that allow soldiers to have real-time maps

Local Coverage Map (February 2026)



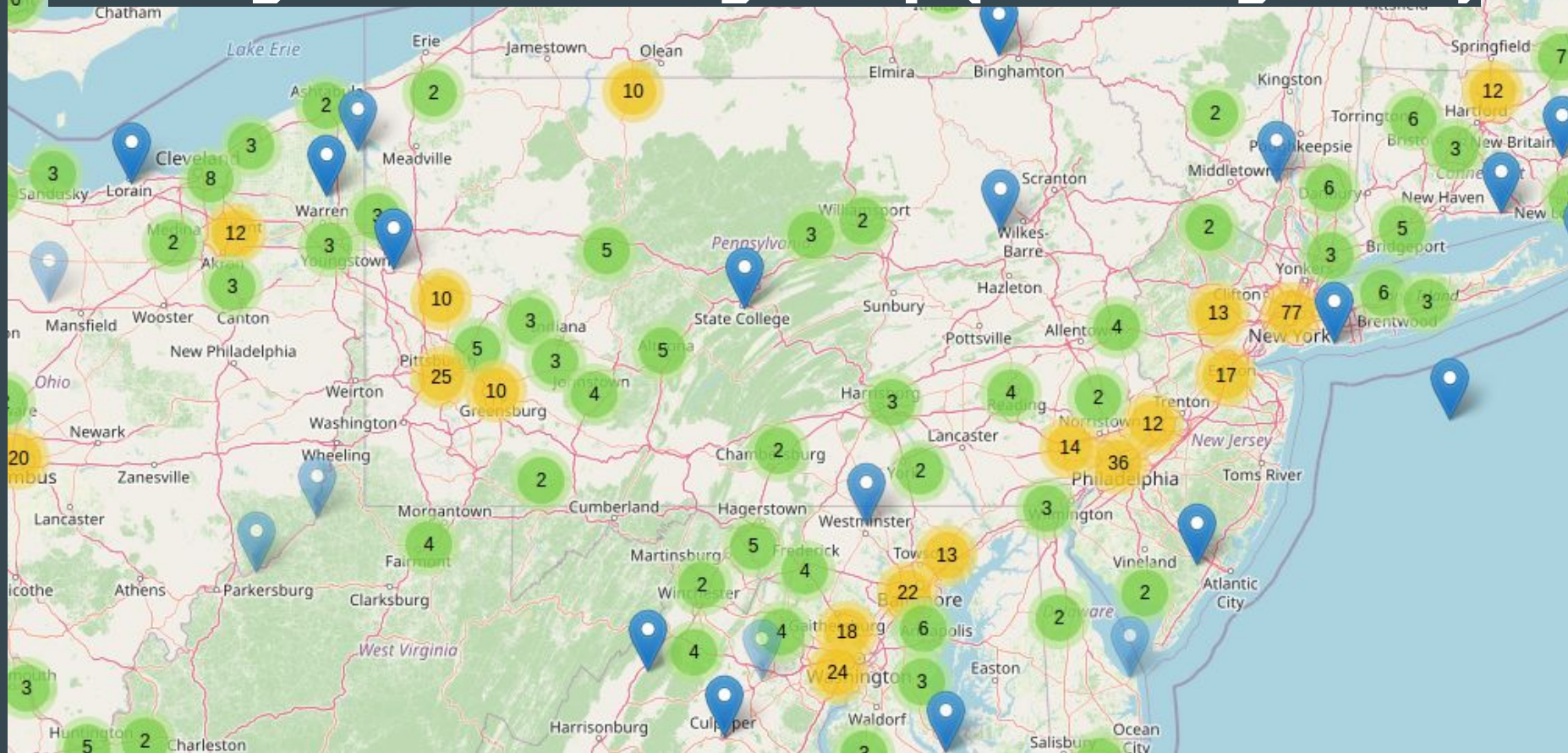
Potential Downsides

- Startup costs, and DIY assembly increase friction for mass adoption
 - Even the cheap devices still cost around \$30
 - Requires line of sight and an established network
 - This technology is still new, and the community is rapidly evolving, but there will be bugs in the meantime
 - Meshtastic's cryptosystem is weaker than Signal's
 - Low bandwidth – can only handle text messages, no voice or image
-
- Bottomline: This isn't a cell phone or Signal replacement, it won't always "just work", but if you practice using it, it can work when nothing else does.

Western Pennsylvania Mesh

- “Best Meshtastic network in the country” - Anthrocon
- 516 nodes
 - 328 as of February (57% growth in 6 months/250% annual growth rate)
- Handles approximately 7000 messages every 24 hours
 - Same as February
- They have a website (wpamesh.net) and a Discord
- 14.6% channel util and 3.36% airtime
 - Channel util is all traffic, airtime is potential useful (human to human) traffic
 - Meshcore utilizes channel much more efficiently than Meshtastic
 - “12% network uptime” on February slides
- 36 routers (18 in February)

Pennsylvania Coverage Map (February 2026)



Other things you can do, from simple to complex

- Check out the 'best mesh in the country' (Anthrocon '26) at <https://wpamesh.net/>
 - On that note, check out the Meshtastic traffic from Anthrocon '26, displayed then and now at <https://awoo.corg.ski/chat>
- Join the [WPAMesh Discord](#) to keep up with the latest and greatest
- Acquire a handheld node and learn how to use it
 - Protomesh [NOMAD](#) open source design (\$30 in parts as of Sep '26)
- Become the Meshtastic expert for your family/friends/community/organization
- Build solar-powered repeater nodes and put them up high
 - Protomesh [HACKERSBLOOM](#) open source design (\$30 in parts as of Sep '26)
 - Free SOLDERDAY workshops every 3rd Saturday 1-3 PM at Prototype x Oakland
- Join the Protomesh Collective to help educate and equip your neighbors in Meshtastic technology

**Questions
?**

Even More Resources!

Low-cost DIY Meshtastic board:

https://github.com/gargomoma/fakeTec_pcb

An honest assessment of the limitations of Meshtastic's cryptosystem:

<https://meshtastic.org/docs/overview/encryption/>

Minneapolis Mesh Map: <https://monitor.mspmesh.org/>

Western Pennsylvania Mesh Map: <https://map.wpamesh.net/map>

Protomesh Collective Resources:

<https://github.com/witchofthewires/protomesh>

Thank You!